Lesson 24: Factorial Designs

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 9.1: Factorial Designs; 9.2: Setting Up a Factorial Experiment; 9.3: Interpreting the Results of a Factorial Experiment; 9.4: Factorial Designs (Summary).

In this lesson

You will learn how factorial designs combine two or more independent variables and how to interpret their three kinds of results: main effects, interactions and simple effects. In English, you will practise the precise wording these results need: *on average*, *depends on*, *but only if*.

Learning goals

After this lesson you should be able to:

- Read a design's notation (IVs, levels, conditions) and draw a factorial design table.
- Choose between- or within-subjects manipulation for each factor, and limit causal conclusions to manipulated variables.
- Identify main effects, spreading and crossover interactions, and simple effects in a table of means.
- Report factorial results accurately with *on average*, *depends on* and *only*.

Key language

Term or phrase	Meaning	Example
factorial design	Each level of one independent variable (IV) is combined with each level of the others to produce all possible combinations; each combination is a condition.	Cell phone use (yes, no) × time of day (day, night)
factorial design table	Rows = levels of one IV, columns = levels of a second IV, each cell = a condition.	A 2 × 2 table has four cells.
mixed factorial design	At least one IV is manipulated between subjects and at least one within subjects.	Phone use within, time of day between
non-manipulated independent variable	A variable (usually a participant variable) treated as an IV but measured, not manipulated.	Private body consciousness
main effect	The effect of one IV averaged across the levels of all other IVs.	Driving was better, on average, without a phone.
interaction	When the effect of one IV depends on the level of another IV.	Disgust mattered only for people high in private body consciousness.

Term or phrase	Meaning	Example
spreading interaction	One IV has an effect at one level of the other IV and a weak effect or no effect at the other level.	A strong phone effect at night, a weaker one by day
crossover interaction	One IV has opposite effects at different levels of another IV.	Caffeine raises extraverts' verbal scores but lowers introverts'
simple effects	The effect of each IV at each level of the other IV; used to break down an interaction.	Caffeine in introverts, then in extraverts
the effect of X depended on Y	Reports an interaction	"the effect of disgust depended on private body consciousness"
but only if	Restricts a finding to one group	"harsher moral judgments—but only if they scored relatively high in private body consciousness"

Rules and explanations

The ideas

Why several IVs? Schnall, Haidt, Clore and Jordan (2008) manipulated disgust (testing in a clean or a messy room) and measured *private body consciousness*, the attention people pay to their own bodily sensations. Participants rated the moral acceptability of behaviours (e.g., eating one's dead dog) from 1 to 7. In the messy room they were more disgusted and judged more harshly, "but only if they scored relatively high in private body consciousness". Interactions like this are "often among the most interesting results in psychological research".

Notation. Each number is one IV (also called a *factor*); its value is that IV's number of levels; their product is the number of conditions: 2 × 2 = 4 conditions, 3 × 2 = 6, 4 × 5 = 20, 2 × 2 × 2 (three IVs) = 8, 2 × 2 × 2 × 3 (four IVs) = 24. More than three IVs is unusual: the conditions become unmanageable, and so does the number of participants needed for adequate statistical power.

Assigning participants. Each IV can be between subjects (simpler, no order or carryover effects, less time per participant) or within subjects (more efficient, controls extraneous participant variables). In the book's mixed design, phone use is within subjects (counterbalanced) and time of day between subjects, so each participant is tested in two of the four conditions.

Non-manipulated IVs. Brown and colleagues (1999) manipulated word type (health-related words such as *tumor* vs. unrelated words such as *election*) and measured hypochondriasis (excessive concern with ordinary bodily symptoms). Such variables are usually participant variables and therefore between-subjects factors. The study is still an experiment if at least one IV is manipulated, but causal conclusions apply only to the manipulated IV. Schnall's team could conclude that disgust affected moral judgments, but not that private body consciousness did: a third variable such as neuroticism might cause both a strict moral code and bodily awareness. If *no* IV is manipulated (measured mood × measured self-esteem), the study is non-experimental.

A worked example. *Invented data for practice*, following the pattern the book describes for its hypothetical driving study (higher = better driving). The table is also the study's factorial design table, with a mean in each cell.

	No cell phone	Cell phone	Mean
Day	8	7	7.5
Night	6	3	4.5
Mean	7	5	

- **Main effect of phone use**: averaged across time of day, 7 vs. 5 (better without a phone).
- **Main effect of time of day**: averaged across phone use, 7.5 vs. 4.5. There is one main effect per IV, and main effects are independent of each other.
- **Interaction**: the phone costs 1 point by day (8 – 7) but 3 at night (6 – 3), so its effect depends on time of day: a *spreading* interaction.
- **Simple effects**: the phone effect by day and at night; the time effect with and without a phone.

With real data, a factorial ANOVA tests whether the main effects and the interaction are statistically significant.

More interactions. Everyday: whether you go to a film may depend on the film *and* on who is coming. Research: Brown's high-hypochondriasis participants recalled more health-related words, but not more neutral words. In a **crossover** (Gilliland, 1980), introverts outperformed extraverts on a verbal test without caffeine, but extraverts outperformed introverts after 4 mg of caffeine per kilogram of body weight; in a line graph the lines literally cross.

Simple effects. An interaction suggests that main effects "may be a bit misleading". In Gilliland's crossover, averaging would wipe out a main effect of caffeine, although caffeine clearly matters; Schnall's main effect of disgust held for only half the participants. A simple effects analysis, needed only when there is an interaction, examines each IV at each level of the other. A 2 × 2 design has 2 main effects and 4 simple effects; 2 × 3 has 2 and 5 (3 + 2); 3 × 3 has 2 and 6.

Language focus

Main effect or interaction? The wording must match the result.

- Main effect: "driving performance was better, *on average*, when participants were not using cell phones than when they were." Signals: *on average*, *overall*, *averaging across*.
- Interaction: "the effect of disgust *depended on* private body consciousness"; "*If* they were high in private body consciousness, *then* those in the messy room made harsher judgments. *If* they were low..., *then* whether the room was clean or messy *did not matter*."

When there is an interaction, do not report a main effect alone; add the restriction (*but only*, *only for*).

> **Note for Arabic speakers** - Arabic has no indefinite article, so *a / an* is often left out. Write "there was *an* interaction", "*a* main effect of disgust", "*a* 2 × 2 factorial design".

Practice

1. Reading questions

1. In Schnall and colleagues' study, which IV was manipulated and how, which was measured, and what was the dependent variable?
2. How many IVs and conditions do a 3 × 3 and a 2 × 2 × 2 design have?
3. Why are non-manipulated IVs "by definition between-subjects factors"?
4. State Brown and colleagues' hypothesis as an interaction.
5. Why might Gilliland's crossover show no main effect of caffeine? Does that mean caffeine has no effect?
6. When is a simple effects analysis necessary?

2. From the book (9.4 Exercises)

1. Practice: "Return to the five article titles presented at the beginning of this section. For each one, identify the independent variables and the dependent variable." (The titles open Section 9.2.)
2. Practice: "Create a factorial design table for an experiment on the effects of room temperature and noise level on performance on the MCAT. Be sure to indicate whether each independent variable will be manipulated between-subjects or within-subjects and explain why." (The MCAT is a long admission test for medical school.)
3. Practice: "Sketch 8 different bar graphs to depict each of the following possible results in a 2 x 2 factorial experiment" (the eight results are listed in 9.4, from "No main effect of A; no main effect of B; no interaction" to "No main effect of A; no main effect of B; interaction"). *Self-study:* for each, first write four cell means (A1B1, A1B2, A2B1, A2B2) and check them; then sketch with A on the *x*-axis, B as two bar colours and the dependent variable on the *y*-axis (the book: "The *y*-axis is always reserved for the dependent variable").

3. Language

1. Combine into one sentence with *depended on*: "People high in hypochondriasis recalled more health-related words than people low in hypochondriasis. The two groups recalled non-health-related words equally well."
2. The book calls this main effect "a bit misleading": "Those in a messy room made harsher moral judgments." Make it accurate.
3. Add the missing articles: "There was interaction between caffeine and personality, which is example of crossover interaction."

Speak with your tutor

- Explain main effects, interactions and simple effects with the book's film example. Ask your tutor to check your use of *on average* and *depends on*.
- Read your eight results from Practice 2.3 aloud as sentences ("There is a main effect of A but no main effect of B, and..."). Ask your tutor to check each against your numbers.

- PSYC 110 Lecture 15 (A Person in the World of People: Morality) presents Jonathan Haidt's moral scenarios, including a family that eats its dead dog, and describes disgust as a basic emotion. Haidt co-authored the Schnall study. Discuss what its factorial design adds to the lecture's point that our moral judgments are governed by factors we might not be conscious of.

Writing task

Your project: a factorial option. Write 220-280 words describing a 2 × 2 factorial version of your study of sleep and memory in university students. For example: your nap condition (a nap opportunity of up to 90 minutes vs. resting quietly for the same time; manipulated) × habitual sleep (short vs. long sleepers, from a one-week sleep diary; measured); dependent variable: word pairs recalled out of 40. If you work on your own topic, combine one manipulated IV with a second IV, manipulated or measured.

Include: the notation; each IV and its levels, saying which is manipulated; between, within or mixed, with reasons; a factorial design table; predicted main effects and interaction; the causal conclusions your design allows; one ethical point (e.g., the confidentiality of sleep diaries).

Self-editing checklist:

- The notation, levels and number of conditions agree.
- I limited causal claims to the manipulated IV.
- I gave a reason for each between/within decision.
- Main effects use *on average* or *averaged across*; the interaction uses *depends on* or *only*.
- I used *a / an* before singular countable nouns (*an interaction*).

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- PSYC 110 Lecture 15, A Person in the World of People: Morality (video): https://oyc.yale.edu/psychology/psyc-110/lecture-15

Answer key

1. Reading questions

1. Disgust was manipulated (a clean or a messy room); private body consciousness was measured with a questionnaire; the dependent variable was the harshness of moral judgments (ratings from 1 to 7) (9.1).
2. 3 × 3: two IVs, nine conditions. 2 × 2 × 2: three IVs, eight conditions (9.2).
3. People are either high or low on them (e.g., hypochondriasis); they cannot be tested in both conditions (9.2).
4. People high in hypochondriasis would recall negative health-related words more accurately than people low in hypochondriasis, but recall non-health-related words about the same (9.3).
5. Averaged across personality, caffeine's positive effect on extraverts would be wiped out by its negative effect on introverts. No: the interaction shows that caffeine's effect depends on personality (9.3).
6. Only when there is an interaction; otherwise "the main effects will tell the complete and accurate story" (9.3).

2. From the book

1. Age and eater type cannot be manipulated; need for closure, a personality characteristic, was probably measured.

Title	Independent variables	Dependent variable
1	temporal delay; orientation	haptic (touch) object recognition
2	intergroup contact; need for closure	prejudice
3	expectancies; coping	pain-induced intentions to smoke
4	age; divided attention	spontaneous recognition
5	food size; package size; restrained vs. unrestrained eater	consumption behaviour

2. (example answer) A 2 × 2 table: room temperature (comfortable vs. hot) in the rows, noise level (quiet vs. noisy) in the columns; four conditions. Both IVs between subjects, with random assignment: taking a long exam four times would cause practice, fatigue and other carryover effects. Other justified answers are acceptable, e.g., noise within subjects with two equivalent test versions in counterbalanced order, to control differences in ability.
3. Example cell means (many others are correct; invented means have no sampling error). A main effect of A: the A means differ. A main effect of B: the B means differ. An interaction: the effect of B at A1 differs from its effect at A2.

Result	A1B1, A1B2, A2B1, A2B2	A1 / A2 means	B1 / B2 means	Effect of B at A1 / A2
1 none	5, 5, 5, 5	5 / 5	5 / 5	0 / 0

Result	A1B1, A1B2, A2B1, A2B2	A1 / A2 means	B1 / B2 means	Effect of B at A1 / A2
2 A	5, 5, 8, 8	5 / 8	6.5 / 6.5	0 / 0
3 B	5, 8, 5, 8	6.5 / 6.5	5 / 8	+3 / +3
4 A, B	4, 6, 7, 9	5 / 8	5.5 / 7.5	+2 / +2
5 A, B, A × B	5, 5, 5, 9	5 / 7	5 / 7	0 / +4
6 A, A × B	4, 6, 8, 6	5 / 7	6 / 6	+2 / −2
7 B, A × B	4, 8, 6, 6	6 / 6	5 / 7	+4 / 0
8 A × B	4, 8, 8, 4	6 / 6	6 / 6	+4 / −4

Results 5 and 7 are spreading interactions; 6 and 8 are crossovers (8 resembles Gilliland's pattern).

3. Language

1. (example answer) The effect of hypochondriasis on recall depended on word type: people high in hypochondriasis recalled more health-related words than people low in hypochondriasis, but there was no difference for non-health-related words.
2. Those in a messy room made harsher moral judgments, but only if they were high in private body consciousness.
3. There was **an** interaction between caffeine and personality, which is **an** example of **a** crossover interaction.

My notes